# H2 Math Summary

Notes

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# 1 Systems of Linear Equations

## 1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

**Example 1.1**

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

**Example 1.2**

The "variables" may be expressions. To find $x$, $y$ and $z$, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-\mathrm{e}^z&=18 && —(1)\\ x^3-\ln y+2\mathrm{e}^z&=9 && —(2)\\ x^3+2\ln y+\mathrm{e}^z&=11 && —(3)\end{aligned}\right\} \Rightarrow x^3=8,\ \ln y=1,\ \mathrm{e}^z=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in $A$, $B$ and $C$. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because $x$ appears as both $x^2$ and $x$, and there are product terms such as $xz$ and $xy$.

## 1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

**Example 1.3**

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for $\lambda$ and $\mu$ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

## 1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

**Example 1.4**

$$\left.\begin{aligned} x+y+z=6 \quad &—(1) \\ x-y+2z=5 \quad &—(2) \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and $z$ can be any real value. So just sub that into $x$ and $y$ to get a correct solution.

$z=-1: (7,0,-1)$ $z=0: (\tfrac{11}{2},\tfrac{1}{2},0)$ $z=1: (4,1,1)$ $z=2: (\tfrac{5}{2},\tfrac{3}{2},2)$

$z=3: (1,2,3)$ $z=4: (-\tfrac{1}{2},\tfrac{5}{2},4)$ $z=5: (-2,3,5)$ $z=6: (-\tfrac{7}{2},\tfrac{7}{2},6)$

We can use the GC table to find all these solutions.

# 2 Inequalities

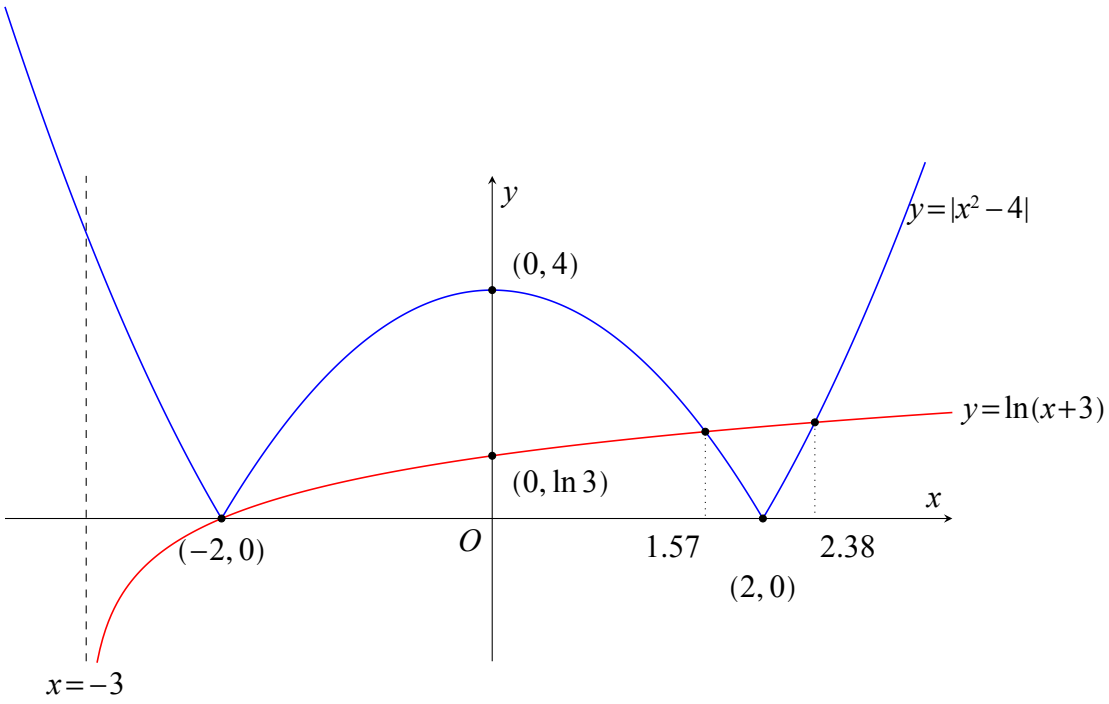
## 2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $f(x) > 0$, or compare $y = f(x)$ with $y = g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the $x$-values where the required graph is above or below.

**Example 2.1**

$$|x^2 - 4| \leqslant \ln(x+3)$$




Using GC, intersections at $x = 1.5746$, $x = 2.3840$

$$\therefore x = -2 \text{ or } 1.57 \leqslant x \leqslant 2.38$$

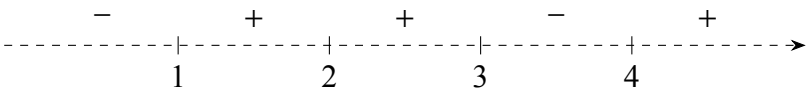
## 2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant $<0$ and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

**Example 2.2**

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$




$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x \leqslant 1$ or $x=2$ or $3<x<4$

You can sketch the graph to check the answer.

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

## 2.3 Common Substitutions

Once an inequality in $x$ is solved, we can replace $x$ with an expression to deduce the solution of a related inequality.

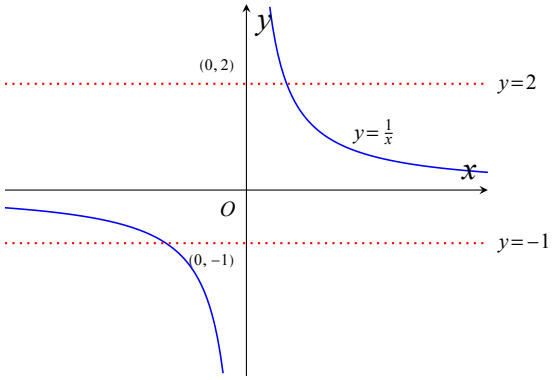
| **Replace $x$ by** | **Original inequality** | | |
|---|---|---|---|
| | $x>2$ | $x<-1$ | $-1<x<2$ |
| $e^x$ | $x>\ln 2$ | no solution | $x<\ln 2$ |
| $\ln x$ | $x>e^2$ | $0<x<e^{-1}$ | $e^{-1}<x<e^2$ |
| $-x$ | $x<-2$ | $x>1$ | $-2<x<1$ |
| $x+1$ | $x>1$ | $x<-2$ | $-2<x<1$ |
| $\lvert x\rvert$ | $x<-2$ or $x>2$ | no solution | $-2<x<2$ |
| $x^2$ | $x<-\sqrt{2}$ or $x>\sqrt{2}$ | no solution | $-\sqrt{2}<x<\sqrt{2}$ |
| $\frac{1}{x}$ | $\left(0<x<\frac{1}{2}\right)$ | $(-1<x<0)$ | $\left(x<-1 \text{ or } x>\frac{1}{2}\right)$ |
| $\cos x,\ 0\leqslant x\leqslant 2\pi$ | (no solution) | (no solution) | $(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$ |

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

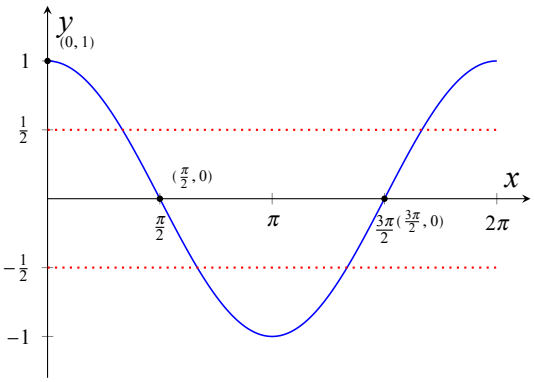
**Example 2.3**

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$




**Example 2.4**

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$
$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

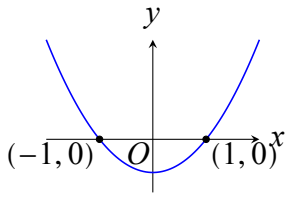


# 3 Graphing Techniques

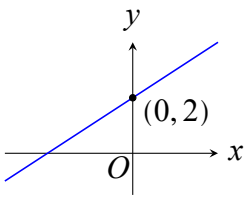
## 3.1 Key Features of a Curve

Key features of a curve:

(a) ***x*-intercepts:** let $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.

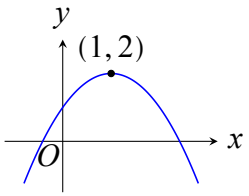



***y*-intercepts:** let $x=0$.
e.g. $y=x+2$: $y=2$.

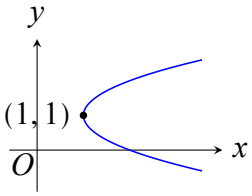



(b) **Stationary points**, when $\frac{dy}{dx}=0$ or $\frac{dy}{dx}$ is undefined. Label upon request.

**Horizontal turning point:** $\frac{dy}{dx}=0$ (tangent horizontal).
e.g. $y=-(x-1)^2+2$, maximum point $(1,2)$.

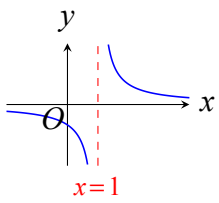
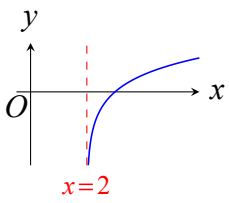
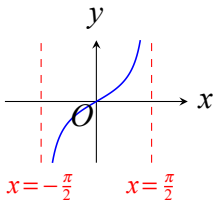



**Vertical turning point:** $\frac{dy}{dx}$ undefined (tangent vertical).
e.g. $x=(y-1)^2+1$, turning point $(1,1)$.



## (c) Asymptotes

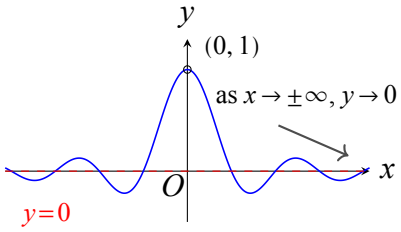
(i) **Vertical asymptotes.** If as $x \to k$, $y \to \pm\infty$, then $x = k$ is a vertical asymptote.

| **Common Occurrence** | **Condition** | **Example and asymptote** | **Graph** |
|---|---|---|---|
| Fractions | denominator $= 0$ | $y = \dfrac{1}{x-1}$<br>$x - 1 = 0 \Rightarrow x = 1$<br>$\therefore x = 1$ is a vertical asymptote |  |
| Logarithmic graphs | inside ln $= 0$ | $y = \ln(x-2)$<br>$x - 2 = 0 \Rightarrow x = 2$<br>$\therefore x = 2$ is a vertical asymptote |  |
| Tangent graphs | inside tan $= \dfrac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$ | $y = \tan x$<br>$x = \pm\dfrac{\pi}{2}, \pm\dfrac{3\pi}{2}, \ldots$ |  |

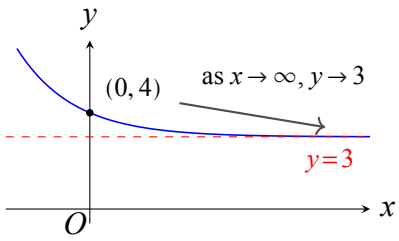
(ii) **Horizontal asymptotes.** As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

Note: there are more cases of horizontal asymptotes, but these are the common cases.



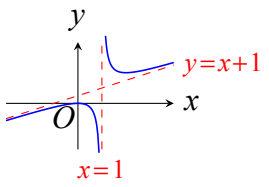
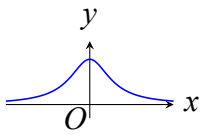
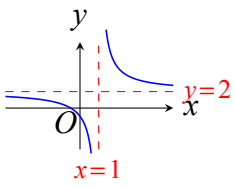

e.g. $y = \dfrac{\sin x}{x}$: curve crosses $y = 0$




e.g. $y = 3 + e^{-x}$

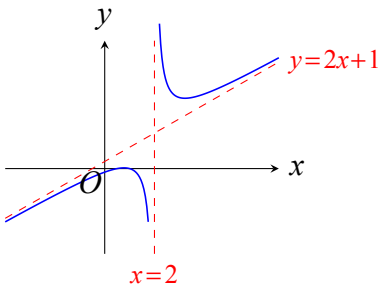
**Useful trick to find horizontal asymptotes**

For rational functions $\frac{P(x)}{Q(x)}$:

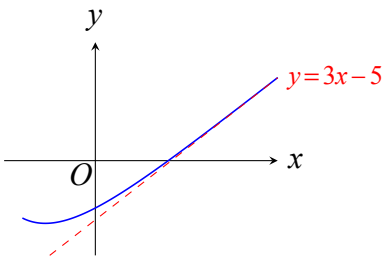
| Case | Horizontal asymptote | Example | Graph |
|---|---|---|---|
| numerator power > denominator power | no horizontal asymptote | $y=\frac{x^2}{x-1}$ |   |
| numerator power < denominator power | $y=0$ | $y=\frac{1}{x^2+1}$ |   |
| numerator power = denominator power | ratio of coefficients of highest power | $y=\frac{2x+1}{x-1}$<br>$y=2$ |   |

(iii) **Oblique asymptotes.** An asymptote of the form $y=mx+c$, $m\neq 0$.

Graphs with oblique asymptotes come in the form $y=mx+c+\mathrm{g}(x)$, where $\mathrm{g}(x)\to 0$ as $x\to\pm\infty$; the oblique asymptote is $y=mx+c$.




e.g. $y=2x+1+\frac{3}{x-2}$



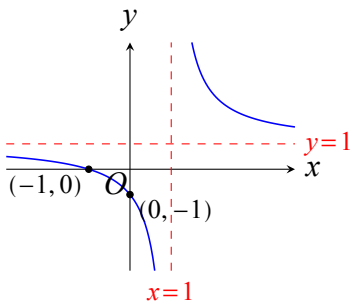
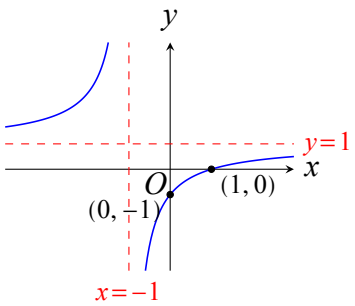
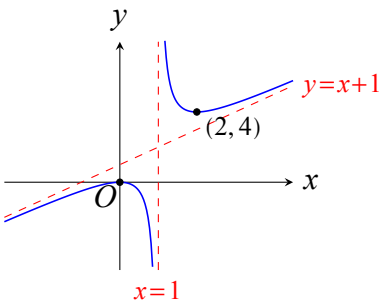
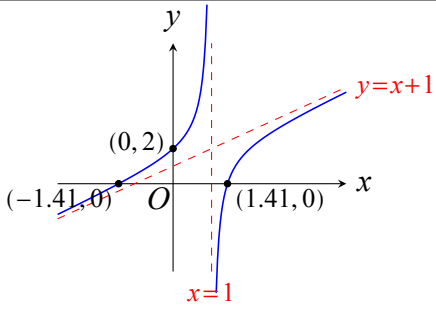
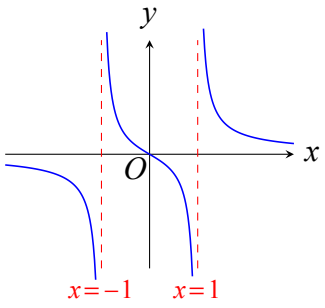
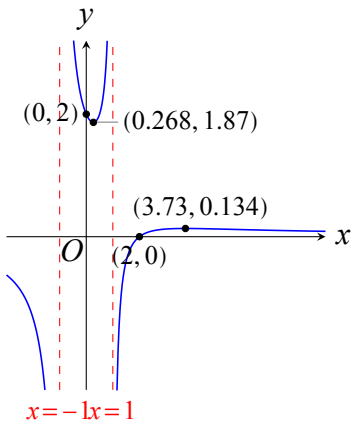

e.g. $y=3x-5+\mathrm{e}^{-x}$

**Note**

For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian equation.

## 3.2 Graphs of Rational Functions

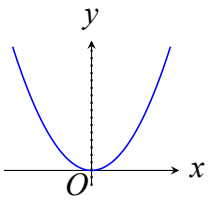
Two common shapes for each type.

| Type and form | Shape 1 | Shape 2 |
|---|---|---|
| **Linear / linear**<br>$y=\dfrac{Ax+B}{Cx+D}$<br>Horizontal asymptote: $y=\dfrac{A}{C}$<br>Vertical asymptote: $x=-\dfrac{D}{C}$ |  <br>$y=\dfrac{x+1}{x-1}$ |  <br>$y=\dfrac{x-1}{x+1}$ |
| **Quadratic / linear**<br>$y=\dfrac{ax^2+bx+c}{dx+e}$<br>$=Ax+B+\dfrac{C}{dx+e}$<br>Oblique asymptote: $y=Ax+B$<br>Vertical asymptote: $x=-\dfrac{e}{d}$ |  <br>2 turning points<br>$y=x+1+\dfrac{1}{x-1}\quad\left(\dfrac{Cd}{A}>0\right)$ |  <br>No turning points<br>$y=x+1-\dfrac{1}{x-1}\quad\left(\dfrac{Cd}{A}<0\right)$ |
| **Linear / quadratic**<br>$y=\dfrac{ax+b}{cx^2+dx+e}$<br>Horizontal asymptote: $y=0$<br>Vertical asymptotes: solve $cx^2+dx+e=0$<br>(possibly no roots) |  <br>$y=\dfrac{x}{(x-1)(x+1)}$ |  <br>$y=\dfrac{x-2}{(x-1)(x+1)}$ |

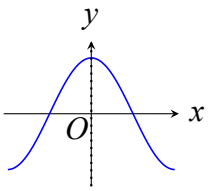
## 3.3 Symmetry

**Even functions:** $f(-x)=f(x)$.

The graph is symmetric about the $y$-axis (line of symmetry $x=0$, dotted).




e.g. $y=x^2$

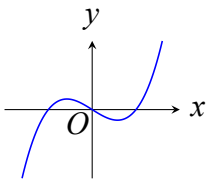



e.g. $y=\cos x$

**Odd functions:** $f(-x)=-f(x)$.

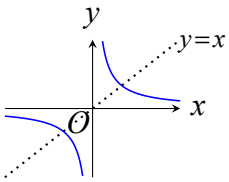
The graph is symmetric about the origin.

e.g. $y=x^3-x$

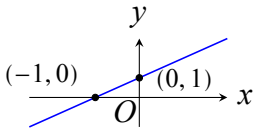
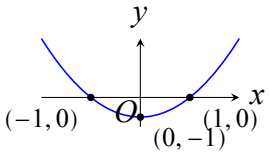
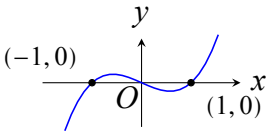
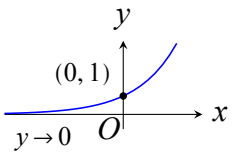
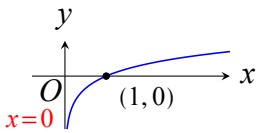
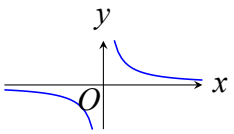
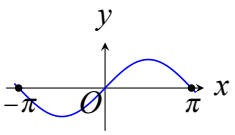
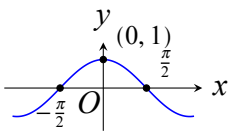
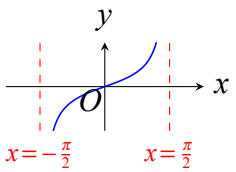
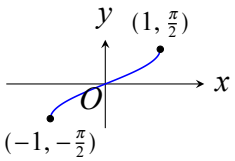
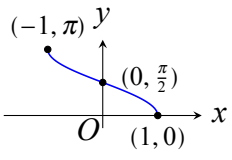
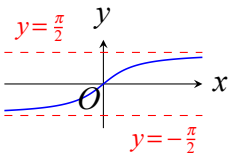



**Symmetry about $y=x$:** swapping $x$ and $y$ gives the same equation.

e.g. $y=\dfrac{1}{x}$ $(xy=1)$




## 3.4 Basic Graphs

| | | |
|---|---|---|
|  Linear $y=x+1$ |  Quadratic $y=x^2-1$ |  Cubic $y=x^3-x$ |
|  Exponential $y=e^x$ |  Logarithm $y=\ln x$ |  Reciprocal $y=\frac{1}{x}$ |
|  $y=\sin x$ |  $y=\cos x$ |  $y=\tan x$ |
|  $y=\sin^{-1}x$ |  $y=\cos^{-1}x$ |  $y=\tan^{-1}x$ |

# 3.5 Parametric Equations

A curve $C$ can be described by giving both coordinates in terms of a parameter $t$:

$$x=x(t), \quad y=y(t), \quad t\in T$$

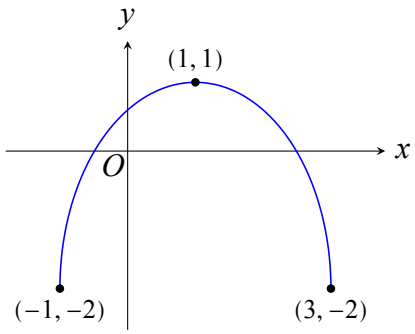
**Example 3.1**

$x=2\cos t+1, \quad y=3\sin t-2, \quad 0\leqslant t\leqslant \pi$

$t=0$: $(3,-2)$ $\quad t=\frac{\pi}{2}$: $(1,1)$ $\quad t=\pi$: $(-1,-2)$

$x=0$: $\left(0, \frac{3\sqrt{3}}{2}-2\right)$

$y=0$: $\left(1\pm\frac{2\sqrt{5}}{3}, 0\right)$




> **CAUTION: sketching from parametric equations**
>
> **ALWAYS take note of the range of $t$ when sketching graphs.** The restriction on $t$ decides which part of the curve exists: above, $0\leqslant t\leqslant \pi$ gives only the upper half of the ellipse, with endpoints $(3,-2)$ and $(-1,-2)$, not the whole ellipse.

## 3.5.1 Finding the Cartesian Equation

The cartesian equation of $C$ is an equation without the parameter $t$. Common techniques:

① **Make $t$ the subject.** e.g. $x=2t-1, y=3t^2+e^t, t\in\mathbb{R}$

$$t=\frac{x+1}{2}$$

$$y=3\left(\frac{x+1}{2}\right)^2+e^{\frac{x+1}{2}}$$

② **Trigo identity formulas.** e.g. $x=2\cos t+1, y=3\sin t-2, 0\leqslant t\leqslant\pi$

$$\cos t=\frac{x-1}{2}, \quad \sin t=\frac{y+2}{3}$$

$$\cos^2 t+\sin^2 t=1$$

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{9}=1, \quad y\geqslant -2$$

③ **Others.** e.g. $x=t-\frac{1}{t}, y=t+\frac{1}{t}, t\neq 0$

$$y+x=2t, \quad y-x=\frac{2}{t}$$

$$(y+x)(y-x)=4$$

$$y^2-x^2=4$$

Note: Avoid finding the cartesian eqn if the question did not ask you to.

## 3.5.2 Asymptotes (Parametric Equations)

- Find (if possible) $t \to t_0$ such that $x \to x_0, y \to \pm\infty$ (vertical asymptote $x = x_0$), or $y \to y_0, x \to \pm\infty$ (horizontal asymptote $y = y_0$).
- Oblique asymptotes (if they exist) can only be found via the cartesian equation.

**Example 3.2**

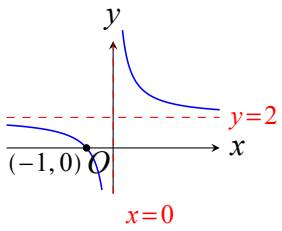
$x = \dfrac{1}{t-1},\ y = 2t,\ t \neq 1$

As $t \to \pm\infty,\ x \to 0,\ y \to \pm\infty$

$\therefore x = 0$ is a vertical asymptote

As $t \to 1,\ x \to \pm\infty,\ y \to 2$

$\therefore y = 2$ is a horizontal asymptote



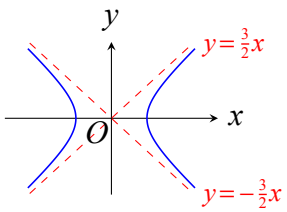

**Oblique asymptotes.** Cannot be found from the parametric form alone; find the cartesian equation first.

e.g. $x = 2\sec\theta,\ y = 3\tan\theta$

$$\sec^2\theta - \tan^2\theta = 1$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

Oblique asymptotes: $y = \pm\dfrac{3}{2}x$




## 3.5.3 Intersections

Intersection between a parametric curve and another curve: sometimes both equations are needed.

e.g. $x = 2t + \ln t,\ y = t^2 - 1,\ t > 0$ and the circle $x^2 + y^2 = 25$:

$$(2t + \ln t)^2 + (t^2 - 1)^2 = 25$$

Using GC, $t = 0.00735$ or $t = 1.86$ (both $> 0$)

$\therefore$ intersections: $(-4.90,\ -1.00)$ and $(4.35,\ 2.47)$

This equation in $t$ cannot be solved algebraically, so the GC is needed.

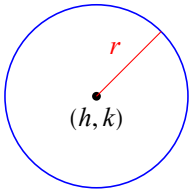
# 3.6 Conics

## Circle

$$(x-h)^2+(y-k)^2=r^2$$

Centre: $(h,k)$

Radius: $r$



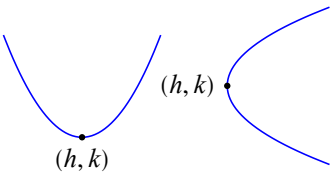

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## Parabola

$$y-k=a(x-h)^2 \qquad x-h=b(y-k)^2$$

$y-k=a(x-h)^2$:
turning point $(h,k)$, $\dfrac{dy}{dx}=0$

$x-h=b(y-k)^2$:
turning point $(h,k)$, $\dfrac{dy}{dx}$ undefined




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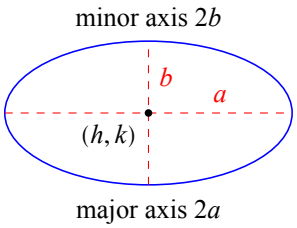
## Ellipse

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Centre: $(h,k)$

"Horizontal radius" $=a$, "vertical radius" $=b$

For $a>b$: major axis horizontal (length $2a$, semi-major $a$); minor axis vertical (length $2b$, semi-minor $b$).




> ⋆ RHS ⇒ numerator coefficients of $x^2$, $y^2$ must all be 1.

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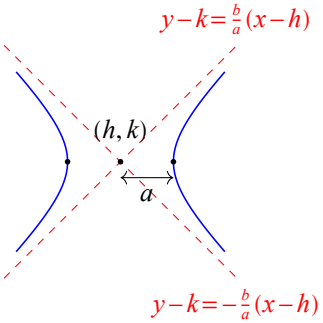
## Hyperbola: Type I

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$$

Centre: $(h,k)$

Distance from centre to vertices $=a$

Asymptotes (dashed): $y-k=\pm\frac{b}{a}(x-h)$




> ⋆ RHS ⇒ numerator coefficients of $x^2$, $y^2$ must all be 1.
> ⋆ Equation of asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

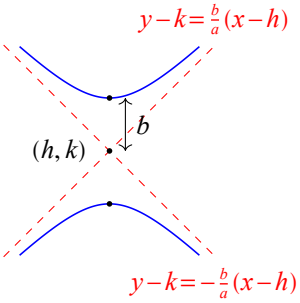
## Hyperbola: Type II

$$\frac{(y-k)^2}{b^2}-\frac{(x-h)^2}{a^2}=1$$

Centre: $(h,k)$

Distance from centre to vertices $=b$

Asymptotes (dashed): $y-k=\pm\frac{b}{a}(x-h)$



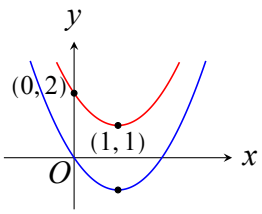
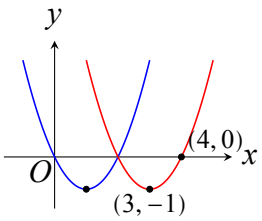
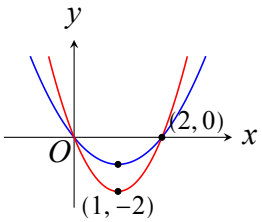
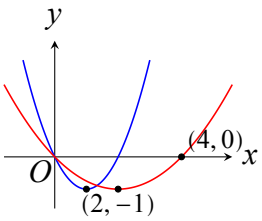

> ⋆ RHS ⇒ numerator coefficients of $x^2$, $y^2$ must all be 1.
> ⋆ Equation of asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

# 4 Graphing Transformations

Every transformation comes from **replacing** $x$ **or** $y$ in the equation $y=\mathrm{f}(x)$.

Throughout, blue is $y=\mathrm{f}(x)=x(x-2)$, with intercepts $(0,0)$, $(2,0)$ and minimum point $(1,-1)$; red is the transformed graph.

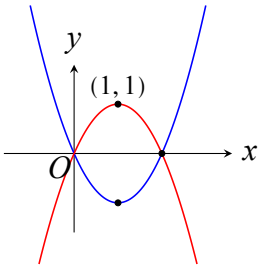
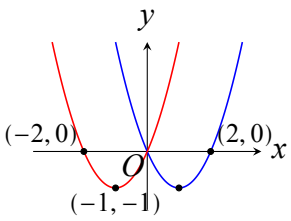
## 4.1 Translations and Scalings

| Transformation | Replacement | Result | Graph |
|---|---|---|---|
| Translation of $a$ units in the positive $y$ direction | replace $y$ with $y-a$: $y-a=\mathrm{f}(x)$ | $y=\mathrm{f}(x)+a$ (graph: $a=2$) |   |
| Translation of $a$ units in the positive $x$ direction | replace $x$ with $x-a$ | $y=\mathrm{f}(x-a)$ (graph: $a=2$) |   |
| Scaling by factor $a$ parallel to the $y$-axis | replace $y$ with $\frac{y}{a}$: $\frac{y}{a}=\mathrm{f}(x)$ | $y=a\mathrm{f}(x)$ (graph: $a=2$) |   |
| Scaling by factor $a$ parallel to the $x$-axis | replace $x$ with $\frac{x}{a}$ | $y=\mathrm{f}\left(\frac{x}{a}\right)$ (graph: $a=2$) |   |

**CAUTION**

Every replacement acts the "opposite" way to what it looks like: $x-2$ moves the graph in the **positive** $x$ direction; $y=\mathrm{f}(2x)$ is a scaling by factor $\frac{1}{2}$ (not 2) parallel to the $x$-axis.
Replacing $x$ does **not** mean replacing the input of f, and replacing $y$ does not mean replacing the output $\mathrm{f}(x)$.
If you are ever confused, think of $\sin x$ vs $\sin 2x$.

## 4.2 Reflections

| Transformation | Replacement | Result | Graph |
|---|---|---|---|
| Reflection about the $x$-axis | replace $y$ with $-y$: $-y=\mathrm{f}(x)$ | $y=-\mathrm{f}(x)$ |   |
| Reflection about the $y$-axis | replace $x$ with $-x$ | $y=\mathrm{f}(-x)$ |   |

## 4.3 Combining Transformations

Functions of the form $y=a\mathrm{f}(bx+c)+d,\ \ a,b,c,d\in\mathbb{R}$: replace $x$ and $y$ **one step at a time**.

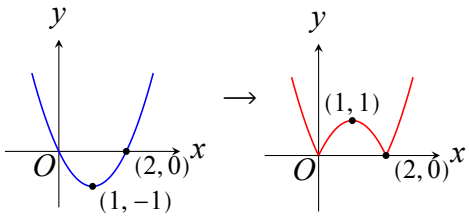
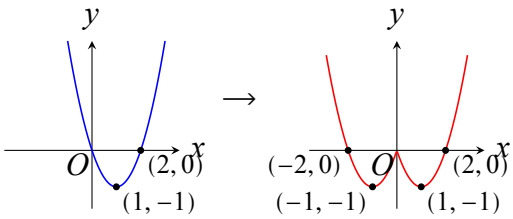
**Example 4.1**

$y=\mathrm{f}(x)\ \longrightarrow\ y=2\mathrm{f}(2x-1)+1$

| Replacement | Transformation | Function |
|---|---|---|
| replace $x$ with $x-1$ | translation of 1 unit in the positive $x$ direction | $y=\mathrm{f}(x-1)$ |
| replace $x$ with $2x$ | scaling by factor $\frac{1}{2}$ parallel to the $x$-axis | $y=\mathrm{f}(2x-1)$ |
| replace $y$ with $\frac{y}{2}$ | scaling by factor 2 parallel to the $y$-axis | $y=2\mathrm{f}(2x-1)$ |
| replace $y$ with $y-1$ | translation of 1 unit in the positive $y$ direction | $y=2\mathrm{f}(2x-1)+1$ |

## 4.4 Modulus Graphs

Every modulus graph can be partitioned into a "+" part and a "−" part: $|x|=x$ for $x\geqslant 0$, $|x|=-x$ for $x<0$.

| Graph | How to sketch | Graph |
|---|---|---|
| $y=\|f(x)\|$ | Keep the part on or above the $x$-axis; reflect the part below the $x$-axis about the $x$-axis.<br>Sharp points at the $x$-intercepts. |  |
| $y=f(\|x\|)$ | Remove the part on the left of the $y$-axis; reflect the part on the right of the $y$-axis about the $y$-axis. |  |

**Example 4.2**

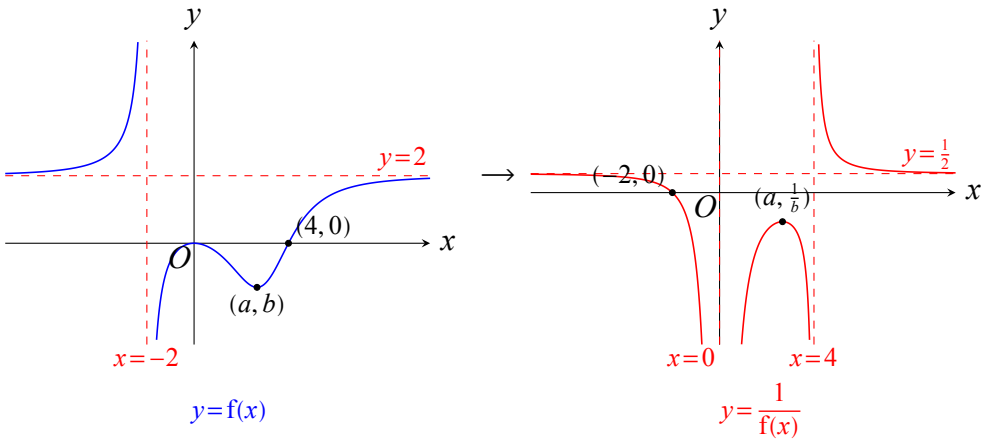
$y=f(x) \longrightarrow y=|f(-2x+1)|$

| Replacement | Transformation | Function |
|---|---|---|
| replace $x$ with $x+1$ | translation of 1 unit in the negative $x$ direction | $y=f(x+1)$ |
| replace $x$ with $-x$ | reflection about the $y$-axis | $y=f(-x+1)$ |
| replace $x$ with $2x$ | scaling by factor $\frac{1}{2}$ parallel to the $x$-axis | $y=f(-2x+1)$ |
| $\|f\|$ | reflect parts below the $x$-axis about the $x$-axis | $y=\|f(-2x+1)\|$ |

## 4.5 Reciprocal Graphs $y=\frac{1}{\mathrm{f}(x)}$

Analyse the new curve at the critical points of $y=\mathrm{f}(x)$.

| | | |
|---|---|---|
| $x$-intercept $(a,0)$ | $\longleftrightarrow$ | vertical asymptote $x=a$ |
| vertical asymptote $x=a$ | $\longleftrightarrow$ | $y\to 0$: point $(a,0)$ (excluded) |
| maximum point $(a,b)$, $b\neq 0$ | $\longleftrightarrow$ | minimum point $\left(a,\frac{1}{b}\right)$ |
| minimum point $(a,b)$, $b\neq 0$ | $\longleftrightarrow$ | maximum point $\left(a,\frac{1}{b}\right)$ |
| horizontal asymptote $y=k$, $k\neq 0$ | $\longleftrightarrow$ | horizontal asymptote $y=\frac{1}{k}$ |
| $y\to\pm\infty$ (incl. oblique asymptote) | $\longleftrightarrow$ | horizontal asymptote $y=0$ |
| $y=\pm 1$ | $\longleftrightarrow$ | same point $(\frac{1}{\pm 1}=\pm 1)$ |
| $\mathrm{f}(x)>0$ / $\mathrm{f}(x)<0$ | $\longleftrightarrow$ | $\frac{1}{\mathrm{f}(x)}>0$ / $\frac{1}{\mathrm{f}(x)}<0$ (same sign) |



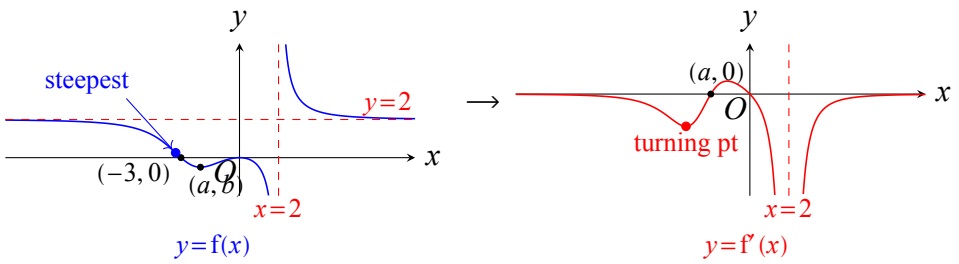

**Note: sketching $y=\frac{1}{\mathrm{f}(x)}$**

- Same sign: $\mathrm{f}>0\Rightarrow\frac{1}{\mathrm{f}}>0$, $\mathrm{f}<0\Rightarrow\frac{1}{\mathrm{f}}<0$.
- $|\mathrm{f}|$ large $\Rightarrow\left|\frac{1}{\mathrm{f}}\right|$ small; $|\mathrm{f}|$ small $\Rightarrow\left|\frac{1}{\mathrm{f}}\right|$ large.
- $\mathrm{f}=0\Rightarrow\frac{1}{\mathrm{f}}$ undefined (vertical asymptote).
- $\mathrm{f}=\pm 1\Rightarrow\frac{1}{\mathrm{f}}=\pm 1$ (graphs meet).

## 4.6 Derivative Graphs $y=\mathrm{f}'(x)$

$\mathrm{f}'(x)=\dfrac{\mathrm{d}y}{\mathrm{d}x}$ is the gradient of $y=\mathrm{f}(x)$: read off the gradient on each interval.

| $y=\mathrm{f}(x)$ | $y=\mathrm{f}'(x)$ |
|---|---|
| stationary point at $x=a$ | $x$-intercept $(a,0)$ |
| increasing / decreasing | above / below the $x$-axis |
| point of inflexion (max. or min. gradient) | turning point |
| vertical asymptote $x=a$ | vertical asymptote $x=a$ |
| horizontal asymptote $y=k$ | horizontal asymptote $y=0$ |
| oblique asymptote $y=mx+c$ | horizontal asymptote $y=m$ |




**i** The turning point on the derivative graph corresponds to the steepest point on the original graph.

**CAUTION**

**i** Mark $x$-coordinates only where they come from $y=\mathrm{f}(x)$: the $y$-values of $y=\mathrm{f}'(x)$ are generally **not** known from the graph of $y=\mathrm{f}(x)$.

## 4.7 Reverse Transformations

If $y=\mathrm{f}(x)$ undergoes a sequence of transformations to give a known $y=\mathrm{g}(x)$, undo the transformations **in reverse order**, each with its opposite replacement.

**Example 4.3**

$y=\mathrm{f}(x)$ undergoes, in order, the following to give $y=6\mathrm{e}^{-x+3}$. Find $\mathrm{f}(x)$.

A: translation of 2 units in the negative $x$ direction

B: reflection about the $y$-axis

C: scaling by factor 3 parallel to the $y$-axis

| Step | Opposite transformation | Replacement | Function |
|---|---|---|---|
| $\mathrm{C}^{-1}$ | scale by factor $\frac{1}{3}$ parallel to the $y$-axis | replace $y$ with $3y$ | $y=2\mathrm{e}^{-x+3}$ |
| $\mathrm{B}^{-1}$ | reflect about the $y$-axis | replace $x$ with $-x$ | $y=2\mathrm{e}^{x+3}$ |
| $\mathrm{A}^{-1}$ | translate 2 units in the positive $x$ direction | replace $x$ with $x-2$ | $y=2\mathrm{e}^{x+1}$ |

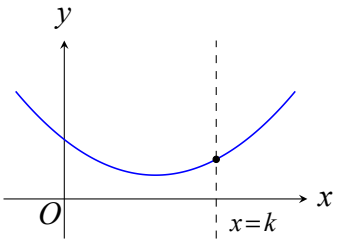
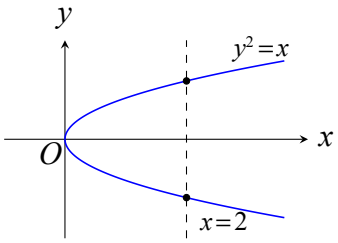
$\therefore\ \mathrm{f}(x)=2\mathrm{e}^{x+1}$

# 5 Functions

## 5.1 Functions, Domain and Range

$$\mathrm{f}: x \mapsto \underbrace{x^2+3x+2}_{\text{rule f}(x)}, \quad x \in \mathbb{R}, \quad \underbrace{x \geqslant 0}_{\text{domain } D_\mathrm{f}}$$

A function is a relation that gives **exactly one** value of $y$ for each $x \in D_\mathrm{f}$.

| **It's a function** | **Not a function** |
|---|---|
|  every vertical line $x=k$, $k \in D_\mathrm{f}$, cuts the graph **exactly once** |  one vertical line ($x=2$) cuts the graph **more than once** |

| **Inequality** | **Set notation** | **Interval** |
|---|---|---|
| $a<x<b$ | $\{x \in \mathbb{R} : a<x<b\}$ | $(a, b)$ |
| $a \leqslant x \leqslant b$ | $\{x \in \mathbb{R} : a \leqslant x \leqslant b\}$ | $[a, b]$ |
| $a \leqslant x<b$ | $\{x \in \mathbb{R} : a \leqslant x<b\}$ | $[a, b)$ |
| $x>a$ | $\{x \in \mathbb{R} : x>a\}$ | $(a, \infty)$ |
| $x \in \mathbb{R},\ x \neq 0$ | $\mathbb{R} \setminus \{0\}$ | $(-\infty, 0) \cup (0, \infty)$ |

CAUTION

To find the range of a function, **ALWAYS** sketch its graph $y=\mathrm{f}(x)$, $x \in D_\mathrm{f}$, and read the range off the graph. **Never** just sub in the endpoints of the domain.

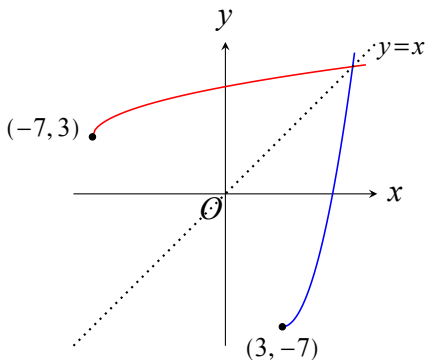
## 5.2 Inverse Functions

| | **Working** |
|---|---|
| Exists | **Exists:** f is one-to-one: every line $y=k,\ k\in\mathbb{R}$, cuts $y=\mathrm{f}(x)$ **at most once** $\Rightarrow \mathrm{f}^{-1}$ exists.<br>$\Rightarrow$ f is one-to-one, thus $\mathrm{f}^{-1}$ exists.<br>**Does not exist:** e.g. $\mathrm{f}(x)=(x-3)^2-7,\ x\in\mathbb{R}$: the line $y=0$ cuts $y=\mathrm{f}(x)$ **twice** (at $x=3\pm\sqrt{7}$) $\Rightarrow \mathrm{f}^{-1}$ does not exist.<br>Alternatively: $\exists\, x_1\neq x_2$ such that $\mathrm{f}(x_1)=\mathrm{f}(x_2)$ $\therefore$ f is not one-one $\Rightarrow \mathrm{f}^{-1}$ does not exist. |
| Rule | make $x$ the subject of $y=\mathrm{f}(x)$ |
| Domain | $D_{\mathrm{f}^{-1}}=R_{\mathrm{f}}$ |
| Range | $R_{\mathrm{f}^{-1}}=D_{\mathrm{f}}$ |
| Graph | reflection of $y=\mathrm{f}(x)$ about $y=x$: all $x$- and $y$-coordinates are swapped, $(a,b)\to(b,a)$ |

**Example 5.1**

$\mathrm{f}: x\mapsto (x-3)^2-7,\ x\geqslant 3$

$$y=(x-3)^2-7$$
$$x-3=\sqrt{y+7}\quad (x\geqslant 3)$$
$$x=3+\sqrt{y+7}$$
$$R_{\mathrm{f}}=[-7,\infty)$$
$$\therefore \mathrm{f}^{-1}: x\mapsto 3+\sqrt{x+7},\ x\in\mathbb{R},\ x\geqslant -7$$




**CAUTION**

- When drawing $y=\mathrm{f}(x)$ and $y=\mathrm{f}^{-1}(x)$ together, use the **same scale** on both axes.
- $\mathrm{ff}^{-1}(x)=\mathrm{f}^{-1}\mathrm{f}(x)=x$, but $D_{\mathrm{ff}^{-1}}=D_{\mathrm{f}^{-1}}=R_{\mathrm{f}}$ while $D_{\mathrm{f}^{-1}\mathrm{f}}=D_{\mathrm{f}}$.
- Solving $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$: it is **not always** $\mathrm{f}(x)=x$ (only when the curves meet on $y=x$).

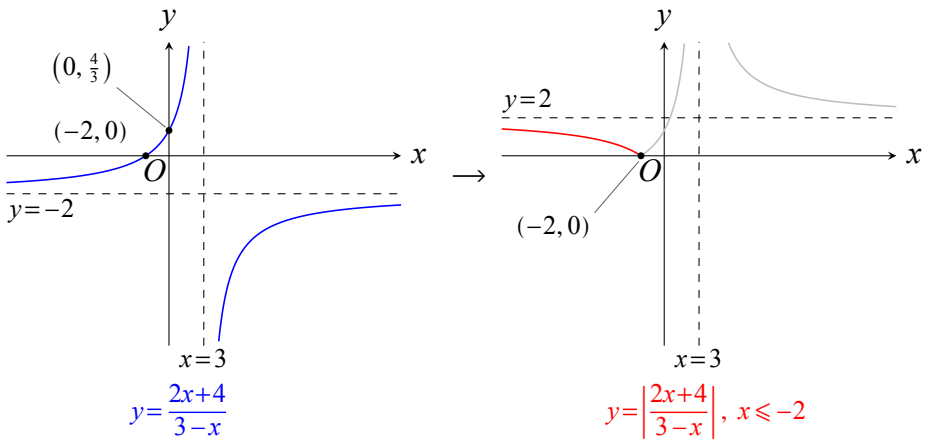
### 5.2.1 Removing the Modulus

On a restricted domain, the sign inside the modulus is fixed, so the modulus can be removed: $|x-a| = x-a$ for $x \geqslant a$, $|x-a| = -(x-a)$ for $x < a$.

**Example 5.2**

$\mathrm{f}: x \to \left|\dfrac{2x+4}{3-x}\right|$, $x \in \mathbb{R}$, $x \leqslant -2$ (2023 A-Level P1 Q7, with the domain restricted to $x \leqslant a$, $a = -2$)

$$y = \mathrm{f}(x) = \left|\frac{2x+4}{3-x}\right|$$
$$= -\frac{2x+4}{3-x} \qquad (x \leqslant -2)$$




**Note**

Whenever you are asked to find $\mathrm{f}^{-1}$ and there is a modulus, you are usually able to remove it in the first step by determining whether the region of the graph is positive or negative for the domain of f.

# 5.3 Composite Functions

| | **Working** |
|---|---|
| Exists | **Exists:** $R_g \subseteq D_f$<br>**Does not exist:** $x_0 \in R_g, x_0 \notin D_f \therefore R_g \not\subseteq D_f$ |
| Rule | sub $g(x)$ into $f(x)$: $f[g(x)]$ |
| Domain | $D_{fg} = D_g$: rightmost function's domain (e.g. $D_{fgh} = D_h$) |
| Range | $D_g \xrightarrow{g} R_g \xrightarrow{f} R_{fg}$: sketch $y = f(x)$ on the domain $R_g$<br>Alternatively, sketch the graph of $fg(x)$ for $D_{fg}$ (avoid, as fg might usually be quite hard to deal with). |

## 5.3.1 Composite with a Piecewise Function

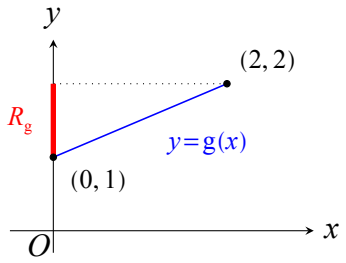
Split according to which piece of f the value $g(x)$ lands in.

**Example 5.3**

$$f(x) = \begin{cases} x^2, & 0 \leqslant x < 1 \\ 2-x, & 1 \leqslant x \leqslant 2 \end{cases} \qquad g(x) = \frac{x}{2} + 1, \ 0 \leqslant x \leqslant 2$$

$$R_g = [1, 2]$$

$$\therefore fg(x) = f(g(x)) = 2 - \left(\frac{x}{2} + 1\right) = 1 - \frac{x}{2}, \quad 0 \leqslant x \leqslant 2$$

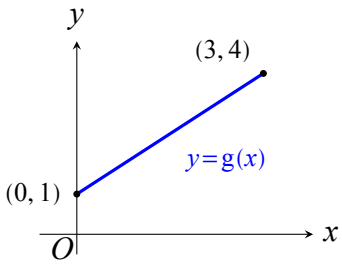


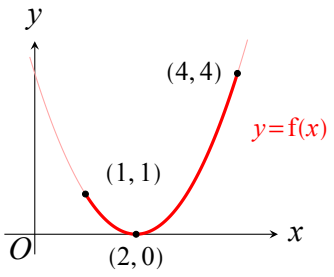

**Example 5.4**

$g(x) = x + 1, \ 0 \leqslant x \leqslant 3; \quad f(x) = (x-2)^2, \ x \in \mathbb{R}$. Assume fg exists.

$$\begin{array}{ccccc} D_g & \xrightarrow{g} & R_g & \xrightarrow{f} & R_{fg} \\ [0, 3] & & [1, 4] & & [0, 4] \end{array}$$

$$\therefore R_{fg} = [0, 4]$$



## 5.4 Functional Powers and Self-Inverse Functions

$\mathrm{f}^2(x)=\mathrm{f}[\mathrm{f}(x)]$ exists if $R_\mathrm{f}\subseteq D_\mathrm{f}$; similarly $\mathrm{f}^3$, $\mathrm{f}^4$, … Note $\mathrm{f}^2(x)\neq[\mathrm{f}(x)]^2$.

**Self-inverse:** $\mathrm{f}=\mathrm{f}^{-1}\iff \mathrm{f}^2(x)=x$ for all $x\in D_\mathrm{f}$ (graph symmetric about $y=x$).

**Periodic composition pattern.** If $\mathrm{g}^2(x)=x$:

$$\mathrm{g}^n(x)=\begin{cases}\mathrm{g}(x), & n \text{ odd}\\ x, & n \text{ even}\end{cases}$$

$$\mathrm{g}^{2027}(x)=\mathrm{g}(x).$$

Similarly, if $\mathrm{f}^3(x)=x$, the pattern repeats every 3: $\mathrm{f}^n=\mathrm{f}^{n \bmod 3}$, and $\mathrm{f}^{-1}=\mathrm{f}^2$.

## 5.5 Piecewise and Periodic Functions

$$\mathrm{f}:x\mapsto\begin{cases}\mathrm{f}_1(x), & x\in D_1\\ \mathrm{f}_2(x), & x\in D_2\end{cases},\qquad \mathrm{f}(x)=\mathrm{f}(x+p)$$

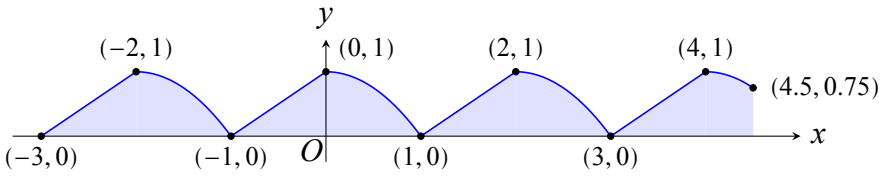
$\mathrm{f}(x)=\mathrm{f}(x+p)$: the graph repeats every $p$ units (period $p$).

→ Always sketch the **original copy** first, then copy & paste.

→ Never sub any $x$ value into f that is outside the domain of the original copy: shift by multiples of $p$ first.

**Example 5.5**

$$\mathrm{f}(x)=\begin{cases}1-x^2, & 0\leqslant x<1\\ x-1, & 1\leqslant x<2\end{cases},\ \mathrm{f}(x)=\mathrm{f}(x+2),\ -3\leqslant x\leqslant 4.5$$




$$\mathrm{f}(4.5)=\mathrm{f}(4.5-4)=\mathrm{f}(0.5)=1-0.5^2=0.75$$

**Area** under a periodic function: area over $n$ full periods $=n\times$ area over one period.

$$\begin{aligned}\int_{-3}^{4.5}\mathrm{f}(x)\,\mathrm{d}x&=\int_{-3}^{3}\mathrm{f}(x)\,\mathrm{d}x+\int_{3}^{4}\mathrm{f}(x)\,\mathrm{d}x+\int_{4}^{4.5}\mathrm{f}(x)\,\mathrm{d}x\\&=3\int_{0}^{2}\mathrm{f}(x)\,\mathrm{d}x+\int_{1}^{2}(x-1)\,\mathrm{d}x+\int_{0}^{0.5}(1-x^2)\,\mathrm{d}x\\&=\frac{107}{24}\approx 4.46\end{aligned}$$

### 5.5.1 Inverse of a Piecewise Function

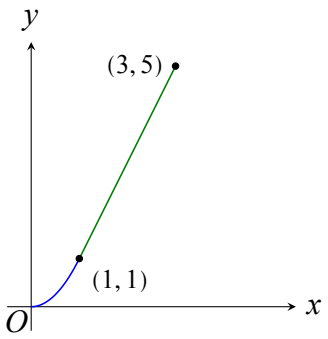
Check one-to-one using the ranges of the pieces (they must not overlap and each piece is one-to-one); then invert each piece, with domain = range of that piece.

**Example 5.6**

$$\mathrm{h}(x)=\begin{cases} x^2, & 0\leqslant x\leqslant 1 \\ 2x-1, & 1<x\leqslant 3 \end{cases}$$

ranges: $[0,1]$ and $(1,5]$ (no overlap)

$$\mathrm{h}^{-1}(x)=\begin{cases} \sqrt{x}, & 0\leqslant x\leqslant 1 \\ \dfrac{x+1}{2}, & 1<x\leqslant 5 \end{cases}$$



# 6 Sequences and Series

A **sequence** is a set of numbers arranged in a definite order, A **series** is the sum of the terms of a sequence:

$u_1,\ u_2,\ u_3,\ \ldots,\ u_n$: the $n$th term

$$S_n = u_1 + u_2 + \cdots + u_n = \sum_{r=1}^{n} u_r, \qquad S_\infty = u_1 + u_2 + u_3 + \cdots = \sum_{r=1}^{\infty} u_r \quad \text{(infinite series)}$$

## 6.1 Recurrence Relations

A recurrence relation defines $u_n$ in terms of one or more preceding terms, together with the initial condition(s). **First-order recurrence relation**: requires only one initial condition; defined using only the immediately preceding term, i.e. $u_{n-1}$ only.

**Second-order recurrence relation**: requires two initial conditions; defined using the two preceding terms, i.e. $u_{n-1}$, $u_{n-2}$.

**Example 6.1**

$$u_{r+1} = \frac{u_r}{2} + 1, \quad u_1 = 4$$

$$u_2 = 3, \quad u_3 = \frac{5}{2}, \quad u_4 = \frac{9}{4}$$

Rule: $u_{r+1} = \frac{u_r}{2} + 1$; initial condition: $u_1 = 4$.

[GC SCREEN PLACEHOLDER] GC 1: generate $u_{r+1} = \frac{u_r}{2} + 1$, $u_1 = 4$ (seq mode, table)

**Example 6.2**

$$u_{r+1} = 2u_r + 1, \quad u_1 = 1$$

**Example 6.3**

$$u_{n+2} = u_{n+1} + 2u_n, \quad u_1 = 1, \quad u_2 = 1$$

$$1,\ 1,\ 3,\ 5,\ 11,\ 21,\ 43,\ \ldots$$

[GC SCREEN PLACEHOLDER] GC: SEQ, $u(n) = u(n-1) + 2u(n-2)$, $u(1) = 1$, $u(2) = 1$ (table)

**Example 6.4**

(GP)

$$u_{n+1} = ru_n, \quad u_1 = a$$

$$a,\ ar,\ ar^2,\ \ldots$$

**Example 6.5**

(AP)

$$u_{n+1} = u_n + d, \quad u_1 = a$$

$$a,\ a+d,\ a+2d,\ \ldots$$

(However, this is not unique.)

$$u_{n+2} - 2u_{n+1} + u_n = 0, \quad u_1 = a, \quad u_2 = a + d$$

## 6.1.1 Limit of a Sequence

$$\text{As } r \to \infty,\ u_r = \frac{1}{r} \to 0 \quad \text{or} \quad \lim_{r\to\infty}(u_r) = \lim_{r\to\infty}\left(\frac{1}{r}\right) = 0$$

$$\text{As } r \to \infty,\ u_r \to L \quad \text{or} \quad \lim_{r\to\infty}(u_r) = L \quad \text{(convergent)}$$

$u_r \to L \Rightarrow u_{r+1} \to L,\ u_{r+2} \to L,\ \ldots$

(For recurrence relations: if the sequence is given to be convergent, change all $u_n$, $u_{n+1}$, $u_{n+2}$ to $L$ and solve for $L$.)

**Rules** ($\lim u_r$, $\lim v_r$ real):

$$\lim_{r\to\infty}(a) = a$$

$$\lim_{r\to\infty}(u_r \pm v_r) = \lim_{r\to\infty}(u_r) \pm \lim_{r\to\infty}(v_r)$$

$$\lim_{r\to\infty}(u_r v_r) = \lim_{r\to\infty}(u_r) \times \lim_{r\to\infty}(v_r)$$

$$\lim_{r\to\infty}\left(\frac{u_r}{v_r}\right) = \frac{\lim_{r\to\infty}(u_r)}{\lim_{r\to\infty}(v_r)}, \quad \lim_{r\to\infty}(v_r) \neq 0$$

## 6.2 Sigma Notation

$$\sum_{r=m}^{n} \mathrm{f}(r) = \mathrm{f}(m)+\mathrm{f}(m+1)+\cdots+\mathrm{f}(n)$$

$$S_n = \sum_{r=1}^{n} u_r = u_1+u_2+\cdots+u_n$$ (By default, $S_n$ starts from $r=1$ unless otherwise stated.)

**Properties:**

$$\text{no. of terms in } \sum_{r=m}^{n} u_r : \; n-m+1$$

$$\sum_{r=m}^{n} a = (n-m+1)a$$

$$\sum_{r=m}^{n} au_r = a\sum_{r=m}^{n} u_r$$

$$\sum_{r=m}^{n} (u_r \pm v_r) = \sum_{r=m}^{n} u_r \pm \sum_{r=m}^{n} v_r$$

$$\sum_{r=m}^{n} u_r = \sum_{r=1}^{n} u_r - \sum_{r=1}^{m-1} u_r$$

(To remember the last three, note that the properties are similar to what you can do in integration.)

## 6.3 Standard Formulas

$$\boxed{\sum_{r=1}^{n} r = \frac{n(n+1)}{2}}$$

AP with first term 1, common difference 1:

$$\sum_{r=1}^{n} r = 1+2+\cdots+n$$

$$S_n = \frac{n}{2}[2(1)+(n-1)(1)]$$

$$= \frac{n(n+1)}{2}$$

$$\boxed{\sum_{r=1}^{n} a^r = \frac{a(a^n-1)}{a-1}, \quad a \neq 1}$$

GP with first term $a$, common ratio $a$:

$$\sum_{r=1}^{n} a^r = a+a^2+\cdots+a^n$$

$$= \frac{a(a^n-1)}{a-1}$$

$$\boxed{\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}}$$

$$\boxed{\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2 = \frac{n^2(n+1)^2}{4}}$$

**Note**

$\sum r^2$ and $\sum r^3$ are given in the exam if needed. All four results hold only if $r$ starts from 1.

## 6.4 Augmented Forms

**Example 6.6**

$$\begin{aligned}\sum_{r=1}^{n}\frac{4}{5^{2r+1}} &= 4\sum_{r=1}^{n}\frac{1}{5\cdot 25^{r}}\\ &= \frac{4}{5}\sum_{r=1}^{n}\left(\frac{1}{25}\right)^{r}\\ &= \frac{4}{5}\cdot\frac{\frac{1}{25}\left[1-\left(\frac{1}{25}\right)^{n}\right]}{1-\frac{1}{25}}\\ &= \frac{4}{5}\cdot\frac{1}{24}\left[1-\left(\frac{1}{25}\right)^{n}\right]\\ &= \frac{1}{30}\left(1-\frac{1}{25^{n}}\right)\end{aligned}$$

> **Note**
>
> To use the $\sum a^{r}$ formula, the power must be exactly $r$, not $r+1$, not $2r$. Rewrite first.

## 6.5 Convergence

**Example 6.7**

Given $u_n=\dfrac{1}{n(n+1)}$ and $S_n=\sum_{r=1}^{n}u_r=1-\dfrac{1}{n+1}$.

| **Convergence of $u_n$:** let $n\to\infty$ in $u_n$. | **Convergence of $S_n$:** let $n\to\infty$ in $S_n$. |
|---|---|
| $u_n=\dfrac{1}{n(n+1)}$<br>As $n\to\infty,\ u_n\to 0$ | $S_n=\sum_{r=1}^{n}\dfrac{1}{r(r+1)}$<br>$=1-\dfrac{1}{n+1}$<br>As $n\to\infty,\ S_n\to 1$ (finite)<br>$\therefore$ the series converges, $S_\infty=1$ |

> **Note**
>
> (1) $u_n\to 0$ is not strong enough evidence that $S_n$ converges, e.g. $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots=\infty$.
> (2) However, if $S_n$ converges, then $u_n$ **must** converge to 0.

# 6.6 Finding $u_n$ or $S_n$ from Each Other

## 6.6.1 Finding $u_n$ from $S_n$

$$u_n = S_n - S_{n-1}, \quad u_1 = S_1$$

**Example 6.8**

$$\begin{aligned} S_n &= n^2 + 2n \\ u_n &= S_n - S_{n-1} \\ &= (n^2 + 2n) - [(n-1)^2 + 2(n-1)] \\ &= 2n + 1 \end{aligned}$$

## 6.6.2 Finding $S_n$ from $u_n$

**Example 6.9**

$$\begin{aligned} u_r &= 3r^2 + r \\ S_n &= \sum_{r=1}^{n} (3r^2 + r) \\ &= 3\sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} r \\ &= 3 \cdot \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \end{aligned}$$